

SUPERVISED LEARNING IN CLINICAL EPIDEMIOLOGY :a review

ABSTRACT

Data science is a deep study of the massive amount of data, which involves extracting meaningful insights from raw, structured, and unstructured data that is processed using the scientific method, different technologies, and algorithms. Data science combines math and statistics, specialized programming, advanced analytics, artificial intelligence (AI) and machine learning with specific subject matter expertise to uncover actionable insights hidden in an organization's data. These insights can be used to guide decision making and strategic planning. Nowadays, many organizations and industries stress using data to improve their products and services. If we talk about just data science, then it is only data analysis using ML Ops machine learning. Both machine learning and data science have to go hand in hand. Engineers have to use ML and data science prominently to make better and more appropriate decisions. Machine learning refers to a series of processes in which a computer finds rules from a vast amount of data. With recent advances in computer technology and the availability of a wide variety of health data, machine learning has rapidly developed and been applied in medical research. Supervised learning approaches to modelling of epidemiologic data are becoming increasingly more prevalent in the literature. These methods have the potential to improve our understanding of health and opportunities for intervention, far beyond our past capabilities.

KEY WORDS-

supervised learning, random forests, gradient-boosted decision tree, support vector machine, neural network

1-INTRODUCTION-

Clinical epidemiology is the study of the patterns, causes, and effects of health and disease in patient populations and the relationships between exposures or treatments and health outcomes.

In recent years, there has been a significant increase in the accessibility of various health care data, such as digital imaging, claims data, registries, and electronic health records. These data are used in scientific study and are linked and integrated [1]. Advances in statistics and computer technology have made it possible to implement intricate and costly algorithms on massive amounts of data. Researchers have tried to create algorithms—known as machine learning—that mimic and even surpass human cognitive ability to do complicated tasks in the medical profession using this data along with technical improvements. Supervised learning in clinical epidemiology is a type of machine learning approach where the algorithm is trained using input-output pairs, or "labelled" data, to predict an outcome. This outcome could be the presence or absence of a disease, the risk of developing a disease, the effectiveness of a treatment, or any other clinical outcome of interest. For example, in a supervised learning model for

predicting heart disease, the input data might include

patient characteristics like age, gender, and cholesterol levels, while the output data might indicate whether the patient has been diagnosed with heart disease or not.

Supervised learning in clinical epidemiology has many potential applications, including risk prediction, treatment prediction, and outcome prediction. It can also be used to identify patterns in large datasets that might not be immediately apparent to human analysts. However, it's important to note that supervised learning models are only as good as the data they're trained on, so it's crucial to ensure that the input data is high-quality and that the model is validated using independent datasets.

2-CONCEPT OF MACHINE LERNING-

In the real world, we are surrounded by humans who can learn everything from their experiences with their learning capability, and we have computers or machines which work on our instructions. But can a machine also learn from

experiences or past data like a human does? So here comes the role of Machine Learning.

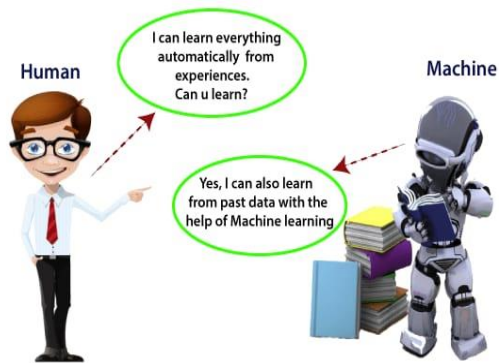


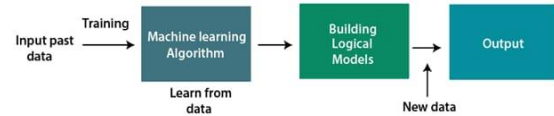
Fig.1

A subset of artificial intelligence known as machine learning focuses primarily on the creation of algorithms that enable a computer to independently learn from data and previous experiences. Arthur Samuel first used the term "machine learning" in 1959. It could be summarized as follows:

- Without being explicitly programmed, machine learning enables a machine to automatically learn from data, improve performance from experiences, and predict things.
- Machine learning algorithms create a mathematical model that, without being explicitly programmed, aids in making predictions or decisions with the assistance of sample historical data, or training data. For the purpose of developing predictive models, machine learning brings together statistics and computer science. Algorithms that learn from historical data are either constructed or utilized in machine learning. The performance will rise in proportion to the quantity of information we provide.

A machine learning system builds prediction models, learns from previous data, and predicts the output of new data whenever it receives it. The amount of data helps to build a better model that accurately predicts the output, which in turn affects the accuracy of the predicted output. Let's say we have a complex problem in which we need to make predictions. Instead of writing code, we just need to feed the data to generic algorithms, which build the logic based on the data and predict the output. Our perspective on

the issue has changed as a result of machine learning. The Machine Learning algorithm's operation is depicted in the following block diagram:



2.1 Supervised Learning, Unsupervised Learning, and Reinforcement Learning.

Supervised machine learning is used for making predictions from data. To be able to do that, we need to know what to predict, which is also known as the target variable. The datasets where the target label is known are called labeled datasets to teach algorithms that can properly categorize data or predict outcomes. Unsupervised learning studies the underlying patterns and predicts the output. In simple terms, in unsupervised learning, the model is only provided with the data in which it looks for hidden or underlying patterns. Unsupervised learning is most helpful for projects where individuals are unsure of what they are looking for in data. It is used to search for unknown similarities and differences in data to create corresponding groups. Unsupervised learning is most helpful for projects where individuals are unsure of what they are looking for in data. It is used to search for unknown similarities and differences in data to create corresponding groups. Reinforcement learning is neither supervised nor unsupervised as it does not require labeled data or a training set. It relies on the ability to monitor the response to the actions of the learning agent. Most used in gaming, robotics, and many other fields, reinforcement learning makes use of a learning agent. A start state and an end state are involved. For the learning agent to reach the final or end stage, different paths may be involved.

3-SUPERVISED LEARNING ALGORITHMS-

Although there is a myriad of algorithms in supervised Learning, the ones frequently used in medical papers are:(i) random forests, (ii) gradient-boosted decision tree (GBDT) (iii) support vector machines (SVM), and (iv) neural networks.

3.1-DECISION TREE AND RANDOM FOREST-

To explain random forests, we will first explain decision trees. A decision tree is a algorithm of

determining the final classification by creating branches from each step(node) based on given rules. Fig.2 share an example of a candidate there is a candidate who has a job offer and wants to decide whether he should accept the offer or not. So, to solve this problem, the decision tree starts with the root node (Salary attribute by ASM). The root node splits further into the next decision node (distance from the office) and one leaf node based on the corresponding labels. The next decision node further gets split into one decision node (Cab facility) and one leaf node. Finally, the decision node splits into two leaf nodes (Accepted offers and Declined offer). Consider the below diagram:

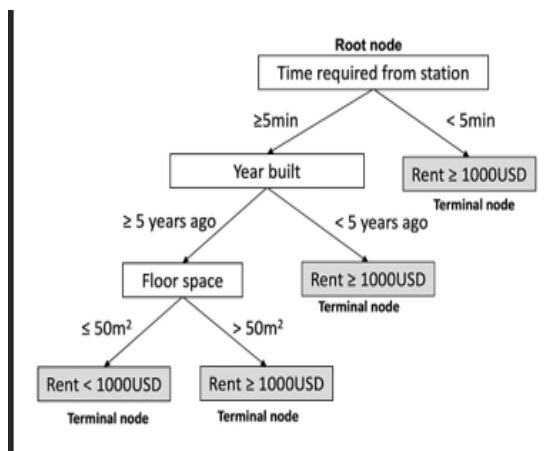


Fig.2

Each rectangle represents a node, of which the first node is called the “root node” and the last node is called the “terminal node”. When splitting a node, the cluttering (impurity) of the node’s data content is expressed as a numerical value called entropy. The node is split to organize its content; that is, the ratio or difference (information gain) of the entropy of the nodes before and after is maximized. Along with entropy, Gini coefficient is also commonly used to measure the impurity value of a split condition. As the tree grows downward, the data can be subdivided according to its characteristics. While complex data can be finely classified, the inherent noise of the data is also captured and classified as a feature. This may cause overfitting; a model that has learned the noise of one particular dataset will not apply well to another new dataset. This is where “random forests” comes in. “Random forests” is a type of ensemble learning (general term for algorithms combining multiple models) that seeks better predictive performance for new data. Random forests create several decision trees and predict by majority vote to prevent overfitting due

to data specific noise (Fig. 3). In random forests, multiple decision trees are created in parallel using randomly selected data for each. Similarly, the variables used to split the decision tree are also chosen at random (). (The variables used in the splitting process are sometimes referred to as “features”.) The name “random forests” is derived from the fact that multiple decision trees, that is, “forests”, are created using “random” data and variables. Random forests can handle complex problems that cannot be represented by linear models at a relatively high speed.

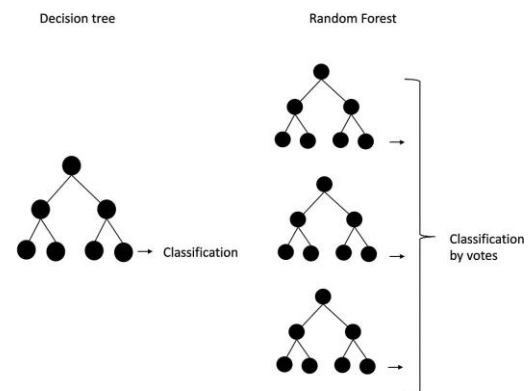


Fig.3

3.2 GRADIENT-BOOSTED DECISION TREE (GBDT)-

GBDT is another type of ensemble learning created by multiple decision trees. While random forests use multiple decision trees created in parallel, GBDT uses multiple decision trees created in sequence. In this algorithm, the first decision tree produces an initial prediction. The second tree predicts the residuals or errors of the first tree by creating another decision tree. Then, the next model predicts the residuals of the precedent model by creating another decision tree, and this process is repeated until the residuals converge to 0 or the number of iterations reaches a prespecified number (i.e., number of decision trees). After the iterations are done, all trees are combined to make a final prediction (Fig.4). GBDT often outperforms other algorithms in terms of accuracy however, it may overfit when the number of trees is too large.

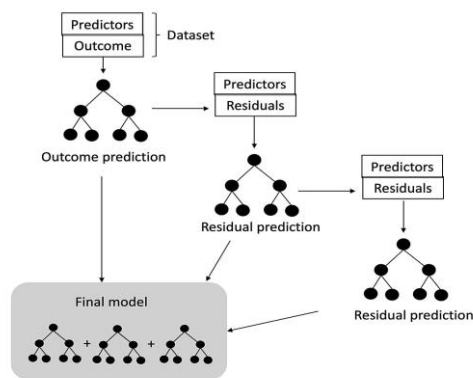


Fig.4

3.3-SUPPORT VECTOR MACHINE (SVM)-

Support Vector Machine or SVM is one of the most popular Supervised Learning algorithms, which is used for Classification as well as Regression problems. However, primarily, it is used for Classification problems in Machine Learning. The goal of the SVM algorithm is to create the best line or decision boundary that can segregate n-dimensional space into classes so that we can easily put the new data point in the correct category in the future. This best decision boundary is called a hyperplane. SVM chooses the extreme points/vectors that help in creating the hyperplane. These extreme cases are called as support vectors, and hence algorithm is termed as Support Vector Machine. SVM classifies data according to a boundary created by an algorithm based on the values of given variables. The model developed by the SVM provide a prediction as to which side of the boundary new data will be on. A clinical application is, for example, to develop a model that predicts death or survival within 30 days based on multiple laboratory test results in a certain disease. Fig.5 illustrates the use of SVM to draw a boundary line that classifies ● and × based on the values of variables X1 and X2 (e.g., laboratory test results). When creating the boundary, the sum of the distances from the boundary to the data should be maximized.

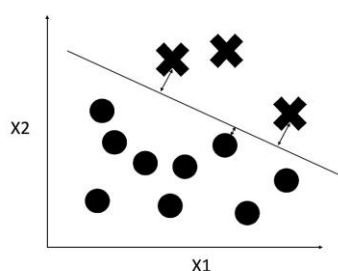


Fig.5

Fig.5 Creating a boundary line using a support vector machine ● and × represent data points of two different groups depicted based on values of variables X1 and X2, respectively (e.g., laboratory test results). The line represents the boundary that maximizes the sum of the distances from the line

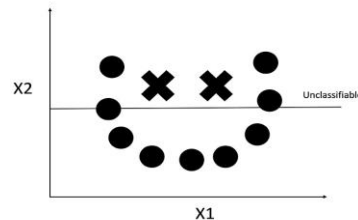


Fig.6

Fig.6 Cases where support vector machine cannot create a linear boundary

● and × represent data points of two different groups depicted based on values of variables X1 and X2, respectively. The linear line cannot separate these data points.

When data points cannot be separated by a linear boundary, as shown in Fig. 6, a method called kernel trick can be used to map the data to a higher dimension called feature space (Fig.7). We can then separate these data points linearly. When the term SVM is used in medical literature, it usually refers to SVM with kernel trick (kernel SVM). Kernel SVM can deal with numerous variables, and it is easy to obtain good results, even with small data. On the other hand, it can be computationally expensive to process a large amount of data as the kernel trick generally increases the dimensionality.

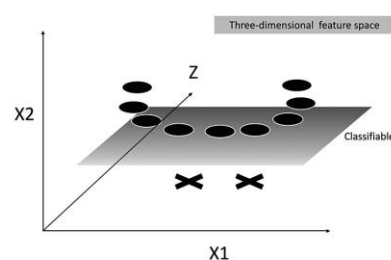


Fig.7

Fig.7 Boundary creation by kernel support vector machine ● and × represent data points of two different groups depicted based on values of variables X1 and X2, respectively. The data are mapped in feature space for linear separation.

3.4-NEURAL NETWORKS IN MACHINE LEARNING-

A neural network is a type of machine learning model that is inspired by the structure and function of the human brain. It is composed of interconnected units called neurons, which work

together to process and interpret complex data. Neural networks are commonly used for tasks such as image recognition, natural language processing, and prediction. They are trained using a large amount of labeled data, allowing them to make accurate predictions on new, unseen data. The network is typically organized in layers: an input layer, one or more hidden layers, and an output layer. Data enters the input layer and flows through the hidden layers, where it undergoes complex transformations, before reaching the output layer to produce a prediction or classification.

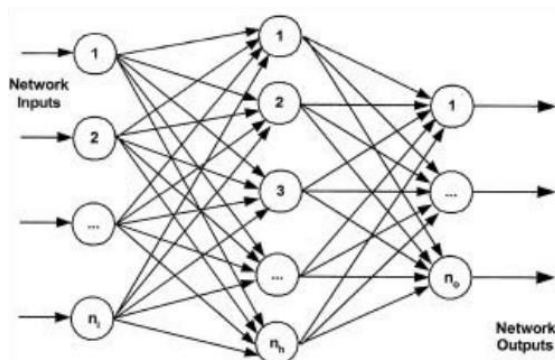


Fig.8

During training, the network is fed large amounts of Labeled data, allowing it to learn patterns and relationships within the data. Through a process called backpropagation, the network adjusts its internal parameters (weights and biases) to minimize the difference between its predictions and the actual labels. This process is repeated many times until the network achieves a satisfactory level of accuracy. Neural networks are highly versatile and can be used for various tasks such as image recognition, natural language processing, and prediction. They have been particularly successful in handling large, unstructured data sets where traditional algorithms struggle. However, they also have some drawbacks, such as being computationally intensive and requiring a large amount of data to train effectively. Overall, neural networks are a powerful tool in the field of machine learning, with numerous applications across different industries and domains.

Neural networks have been widely used in machine learning for a wide range of applications. Some recent use cases include Natural Language Processing (NLP) Neural networks have significantly improved the accuracy of NLP tasks such as sentiment analysis, machine translation, and named entity recognition. Neural networks can be

used to build predictive models for various clinical outcomes, such as patient survival rates, disease progression, and response to treatment. By analyzing large datasets that include patient demographics, clinical variables, and other relevant information, neural networks can help identify factors that contribute to health outcomes.

4-DEVELOPING A PREDICTION MODEL USING SUPERVISED LEARNING-

Developing a prediction model using supervised learning involves several steps. Here's a general outline of the process:

Define the Problem: Before diving into machine learning, it's crucial to understand the problem you're trying to solve. This involves determining what features (input variables) you have and what you're trying to predict (the target variable). For example, if you're predicting house prices, you might have features like size, location, and number of bedrooms, and you want to predict the price.

Collect and Prepare Data: Once you've defined the problem, you need to gather a dataset that contains your input variables and the target variable. This might involve collecting data from sources like real estate websites or databases. You'll also need to clean the data by handling missing values, outliers, and encoding categorical variables if necessary. After cleaning, you'll typically split the data into a training set and a test set.

Feature Engineering: With your cleaned data, you can start creating new features from your existing features. This might include transformations (e.g., log transformation), interactions (e.g., multiplying two variables), or aggregations (e.g., taking the mean of a variable). Feature engineering is crucial because it can help your model make better predictions.

Choose a Model: Once you have your features, you'll need to decide on a model that you think will work well for your problem. This could be a simple linear regression model, a decision tree, a random forest, a neural network, or something else. The choice of model will depend on the problem you're solving, the nature of your data, and the complexity of the relationships you're trying to capture.

Train the Model: With your chosen model, you can start training it on your training set. This involves finding the values for the model's parameters that

minimize the error between the model's predictions and the actual values of the target variable. The goal of training is to teach the model to make accurate predictions on new, unseen data.

Evaluate the Model: Once you've trained your model, you'll want to evaluate its performance on your test set. This could involve using metrics like accuracy, precision, recall, F1 score, or mean squared error, depending on the problem you're solving. The goal of evaluation is to ensure that your model is making accurate predictions and to identify any areas where it might be underperforming.

Hyperparameter Tuning: With your model's performance evaluated, you may want to fine-tune its hyperparameters to optimize its performance. This could involve using techniques like grid search, random search, or Bayesian optimization. The goal of hyperparameter tuning is to find the best combination of hyperparameters that results in the best performance on your test set.

Deploy the Model: Once you're happy with your model's performance, you can deploy it to make predictions on new, unseen data. This could involve creating an API, integrating it into a web application, or some other form of deployment. The goal of deployment is to make your model accessible and usable by others who can benefit from its predictions.

Monitor and Maintain the Model: After deploying your model, you'll need to keep an eye on its performance in production. This might involve monitoring its predictions, retraining it periodically with new data to keep it accurate, or making adjustments to the model based on feedback from users. The goal of monitoring and maintenance is to ensure that your model remains accurate and useful over time.

Interpret the Model: Finally, it's important to try to understand how your model is making its predictions. This could involve looking at feature importances, examining the model's coefficients, or using techniques like SHAP or LIME to explain individual predictions. The goal of interpretation is to gain insights into the underlying relationships in your data and to understand how your model is using those relationships to make predictions.

5-SUPERVISED LEARNING APPLICATIONS-

Example 1: Triage Systems Developed by Multiple Machine Learning Methods

Where medical resource is limited, differentiation and prioritization of critically ill children are important. Goto et al. [37] examined how well an objective triage system developed by machine learning can predict clinical outcomes of children presented to the emergency department (ED) compared to a conventional triage method based on a medical professional's assessment. The authors predicted in-hospital death and/or ICU admission using the least absolute shrinkage and selection operator or lasso in short, random forest, GBDT, and deep neural network. Variables used for prediction were age, sex, mode of arrival (walk-in vs ambulance), vital signs (temperature, pulse rate, systolic and diastolic blood pressure, respiratory rate, and oxygen saturation), visit reasons, patient's residence (home vs other [e.g., long term care facility]), ED visits in the preceding 72 hours, and comorbidities. All the machine learning-based triage systems, although not statistically significant, performed better than the conventional triage system with a fewer number of under triaged children. The authors concluded that machine learning-based triage systems may support clinicians in making triage decisions efficiently, thus improving optimal resource allocation.

5.2 <Example 2> Prognostic Prediction for COVID-19 Using a Combination of Machine Learning Methods

Health care systems worldwide are overwhelmed by the soaring number of COVID-19 patients. For early intervention and optimal resource allocation, an accurate prediction model for COVID-19 is needed. In contrast to Goto et al. [37] in example 1 where they evaluated multiple models separately, Gao et al. [38] integrated four different machine learning models into one ensemble model to predict the mortality risk of admitted patients for COVID-19. To develop the ensemble model, the authors first selected important 14 variables (consciousness, male, age, sputum, blood urea nitrogen, respiratory rate, D dimer, number of comorbidities, platelet count, fever, albumin, SpO₂, lymphocyte, and chronic kidney disease) out of original 53 variables by lasso, another machine learning method often used for unimportant variable elimination. With

the 14 variables, the authors developed 6 machine learning prediction models (logistic regression, SVM, GBDT, neural network, k-nearest neighbor, and random forests), then integrated the top 4 predictive models (logistic regression, SVM, GBDT, and neural network) into one. The ensemble model achieved an area under the curve of 0.96 and 0.92 for predicting mortality of COVID-19 in two external cohorts. The authors concluded that the model efficiently enables accurate risk stratification of COVID-19 patients on admission.

6-CHALLENGES FACE IN MACHINE LEARNING-

Supervised learning in machine learning is a powerful approach used in various applications. However, it also faces several challenges that need to be addressed for successful implementation. One of the main challenges is the availability and quality of labeled data. Labeled data is essential for training supervised learning models, but it can be difficult and expensive to obtain, especially for niche or specialized domains. Additionally, labeled data may be prone to errors or biases, which can affect the performance of the trained models. Another challenge is the trade-off between model complexity and generalization. More complex models can learn intricate patterns in the data, but they are also more prone to overfitting, where the model performs well on the training data but poorly on unseen data. On the other hand, simpler models may not capture the full complexity of the data and may underfit, performing poorly on both training and unseen data. Addressing these challenges requires careful consideration of the data, model selection, and tuning of hyperparameters to strike the right balance between model complexity and generalization. One of the challenges in supervised learning is the "curse of dimensionality." As the number of input features increases, the size of the feature space grows exponentially. This can lead to increased computational complexity, requiring larger amounts of data to train effectively. Another challenge is dealing with "unbalanced data." If one class is significantly underrepresented in the data, the model may not learn to accurately predict it. In many cases, the "black box" nature of some complex models, like deep neural networks, can be a challenge. These models are often difficult to interpret, making it hard to understand how they make predictions. Moreover, ensuring that a model can perform well on new, unseen data that may have slightly different characteristics than the training data is essential for real-world applications.

This is known as the challenge of "generalization." Finally, addressing these challenges often requires a combination of domain knowledge, algorithmic improvements, data preprocessing, and careful experimental design. Balancing these considerations can be a significant challenge in supervised learning.

7-CONCLUSION-

The integration of machine learning into restorative medicine requires intricate collaboration between computer researchers, analysts, healthcare experts, and space specialists. Collaborative endeavors can offer assistance recognize pertinent information sources, create vigorous calculations, translate demonstrate yields, and guarantee the moral and capable utilize of machine learning innovations in healthcare settings. Besides, continuous inquire about is required to address basic challenges such as information protection, interpretability of machine learning models, calculation inclination, and the generalizability of discoveries over different understanding populaces. By tending to these challenges, we will open the complete potential of machine learning to progress persistent care, optimize treatment methodologies, and progress restorative information.

In conclusion, whereas machine learning holds incredible guarantee for revolutionizing pharmaceutical, its fruitful application requires a nuanced understanding of the complexities inherent in restorative information and clinical decision-making. By cultivating collaboration, development, and dependable usage, able to use the complementary qualities of machine learning and human ability to realize transformative progressions in healthcare.

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